

RAKESH KUMAR MANDAL

+91 88266 73734 | rakeshmandaltech@gmail.com | LinkedIn | GitHub

Education

Manav Rachna International Institute of Research and Studies <i>Bachelor of Technology (B.Tech), Computer Science Engineering</i>	<i>Faridabad, Haryana</i> 2023 – 2027
Govt. Co-Ed Sr. Sec. School, Lajpat Nagar <i>Senior Secondary (Class XII)</i>	<i>New Delhi</i> 2023
New Bal Vaishali Pub School, Meethapur <i>Secondary (Class X)</i>	<i>New Delhi</i> 2021

Projects

AI Resume Builder *React.js, Node.js, Express.js, MongoDB, Tailwind CSS, Gemini AI, ImageKit*

- Architected a full-stack web application using the MERN stack that allows users to seamlessly create, customize, and download professional resumes as PDFs.
- Integrated Gemini API to provide dynamic, AI-driven content enhancement, assisting users with intelligent phrasing and bullet-point generation.
- Engineered a secure backend architecture with Node.js and Express, implementing JWT and Bcrypt for robust user authentication and data protection.
- Designed a highly responsive UI with Tailwind CSS and managed complex application state using Redux Toolkit, supporting multiple templates and custom accent colors.

Seat Belt Detection using Deep Learning *Python, YOLOv11, OpenCV*

- Developed a real-time seat belt detection system using YOLOv11 and OpenCV for automated driver safety monitoring in varied driving conditions.
- Achieved strong model performance with **mAP of 94.6%**, precision of 95.3%, and recall of 90.0% on a custom-labeled test dataset.
- Implemented a full pipeline covering data preprocessing, model inference, and bounding box visualization for accurate real-time detection.
- Published findings as a research paper analyzing object detection performance of YOLOv11 on safety-critical applications.

Restaurant Website *HTML, CSS, JavaScript*

- Designed an interactive restaurant website featuring a dynamic menu display, smooth scrolling navigation, and a contact section.
- Implemented JavaScript-driven features including image sliders and animated transitions for an engaging user experience.
- Optimized layout for both mobile and desktop devices using responsive CSS, achieving fast load times and clean visual design.

Certifications & Achievements

- **Research Paper Presentation (ICASET 2025)** — Presented a research paper on seat belt detection using YOLO models at the 4th International Conference on Advances in Science, Engineering & Technology (ICASET).
- **Cisco Networking Foundation Certification** — Completed coursework covering networking fundamentals, IP addressing, and network security principles.
- **HackWithDelhi Hackathon** — Participated in a competitive hackathon, demonstrating problem-solving and project development skills.

Technical Skills

- **Languages:** C, C++, JavaScript, Python
- **Frameworks:** React.js, Next.js, Tailwind CSS, Redux Toolkit
- **Data Science:** NumPy, Pandas, Matplotlib, OpenCV
- **Full-Stack:** MongoDB, Express.js, Node.js, HTML, CSS, REST APIs
- **Core Concepts:** DSA, OOP, DBMS, Computer Networks
- **Tools:** Git, GitHub, Google Colab, VS Code, ImageKit